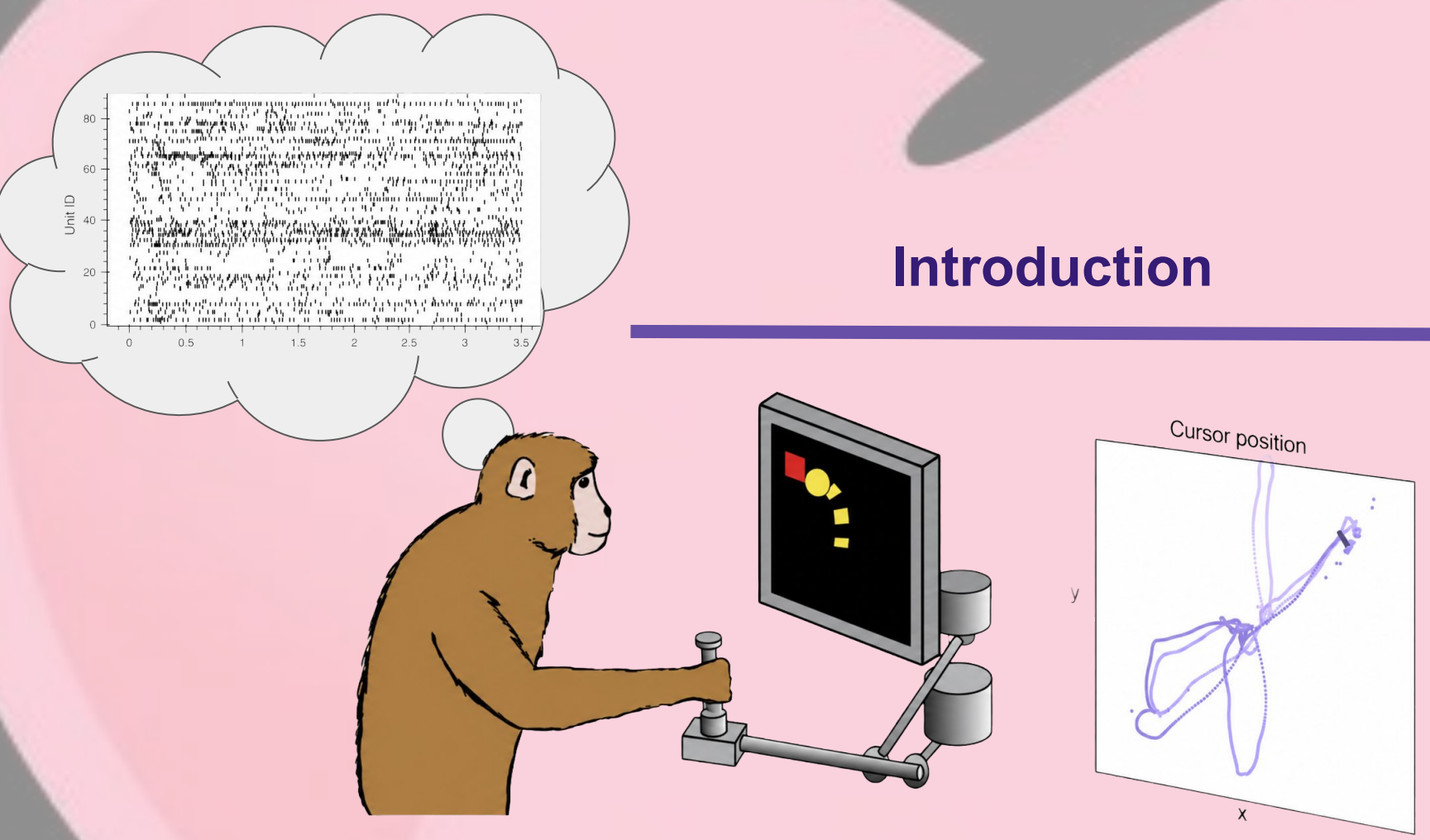


Pre-training On manY neurOns POYO!

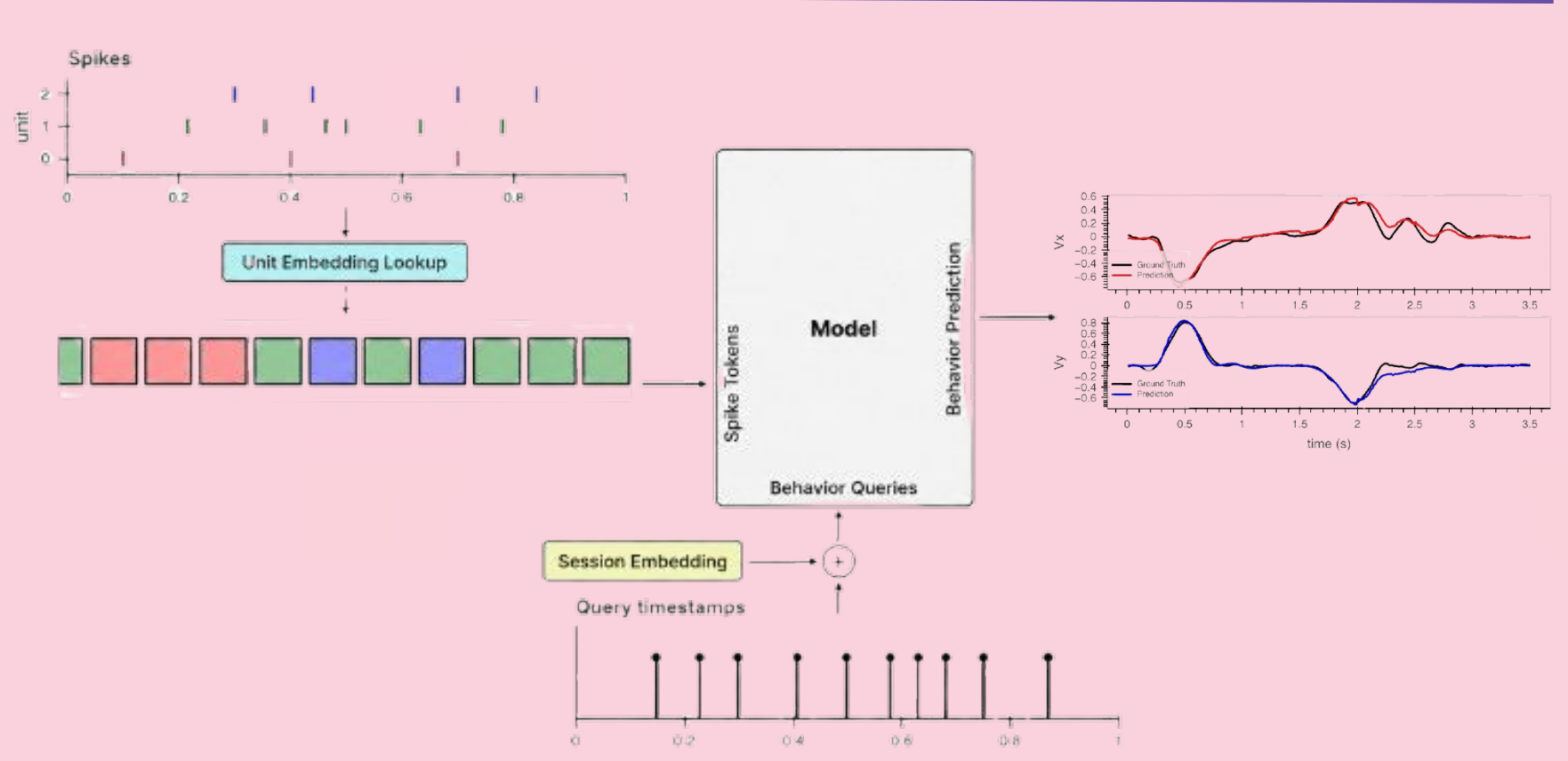
Benjamin Mun¹, Kaden Priebe¹
¹Cornell University

Introduction

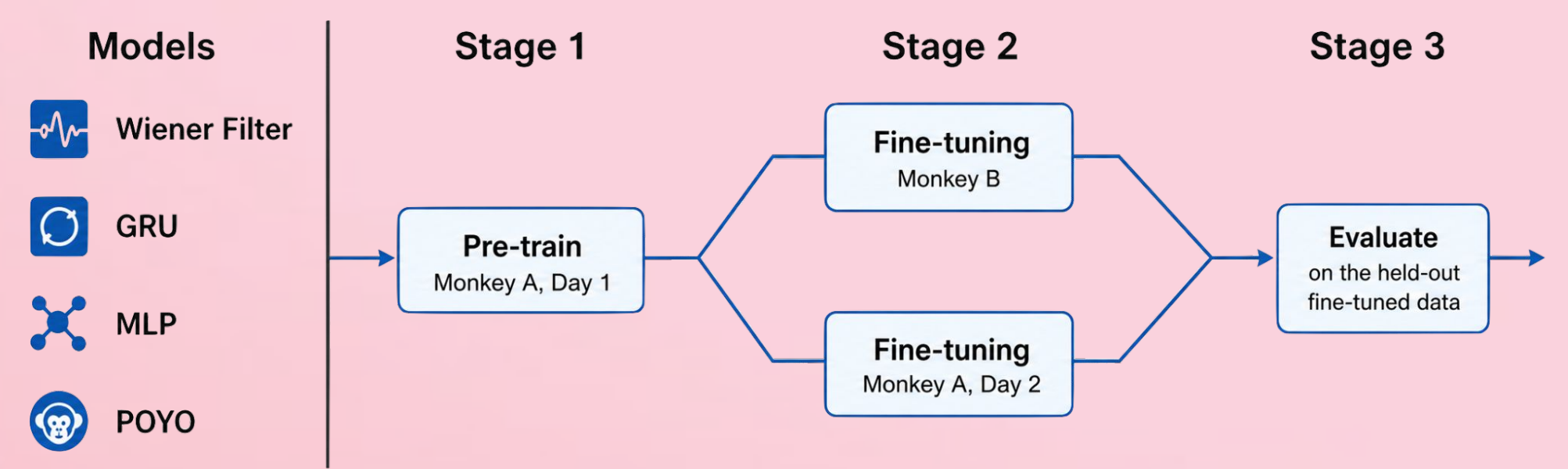


Challenge: Model trained on one lab session does not generalize to a different lab session

Architecture: POYO



Methodology



Independent Study

- What would happen to POYO if tokenization changed?

Result

Models	Same Animal, New Day	New Animal
Wiener Filter	0.21673	0.21673
GRU	0.609909	0.614182
MLP	0.677891	0.672641
POYO	0.707492	0.712334
POYO-binned	0.318293	0.261797

Conclusion

- POYO generalizes to a new animal and new day neural data better than previous models
- POYO's unique tokenization scheme in contrast to the industry standard boosts its performance

Future Direction

- Scalability
- Zero-shot capability
- Multimodal
- Real Time Decoding Speed

References

